Amanjot Singh

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SUMMARY

- Full-stack software developer with 5 years of professional experience designing, building, testing, and deploying enterprise web applications, reusable backend services, REST APIs, and automation solutions
- Production experience with Java, Spring Boot, React, TypeScript, Python, and MSSQL, running containerized services with Docker and Kubernetes on Azure and shipping through GitHub Actions CI/CD pipelines
- Strong background in Agile development, the SDLC, automated unit and integration testing, root-cause debugging, code reviews, and technical documentation in collaborative teams
- Pursuing a B.S. in Computer Science (AI & ML stream) at Carleton University with an 11.42/12 GPA, and building LLM-powered applications with the Claude API

TECHNICAL SKILLS

- **Languages:** Java, TypeScript, JavaScript, Python, SQL, Bash/Shell, C++, C, HTML, CSS, ColdFusion
- **Backend:** Spring Boot, Node.js, Express.js, REST APIs, Microservices, API Authentication, JSON, XML
- **Front-end:** React, TypeScript, HTML/CSS, Bootstrap, jQuery, Responsive Web Design, Qt
- **Databases:** MSSQL, PostgreSQL, MySQL, SQLite, MongoDB, Elasticsearch, Relational Database Design, ORM
- **Cloud & DevOps:** Docker, Kubernetes, Azure, Azure Automation, Microsoft Power Platform, AWS (S3, EC2), GitHub Actions, CI/CD Pipelines
- **Linux & Debugging:** Linux (Ubuntu, Fedora), Windows, WSL, systemctl, gdb, valgrind, perf, vim
- **Testing & Practices:** Unit & Integration Testing, Test Automation, Debugging, Code Reviews, Agile/SDLC, Object-Oriented Design, Technical Documentation, Git Workflows
- **AI Tools:** Claude, Claude API, GitHub Copilot, Cursor, ChatGPT, AI-Assisted Development
- **Tools:** Git, GitHub, Postman, IntelliJ IDEA, Eclipse, JetBrains Suite, Tableau, Power BI

WORK EXPERIENCE

Web Application Developer **Oct 2021 – Present**

National Judicial Institute *Ottawa, ON*

- Design, build, and maintain full-stack enterprise applications and reusable backend services in Java and Spring Boot exposing REST APIs, powering digital learning platforms used by thousands of Canadian judges, with a focus on reliability, maintainability, and documentation
- Develop front-end features in React and TypeScript, working across the full stack from UI through API to database
- Investigate and resolve production issues across front-end, backend, database, and external integration layers, following each from diagnosis through root cause to a tested, deployed fix
- Improve application performance through database query optimization, API improvements, caching strategies, and front-end rendering optimizations
- Build CI/CD pipelines with GitHub Actions and workflow automations using Azure Automation, Microsoft Power Platform, and Python/Bash scripts to streamline production deployments and eliminate manual tasks, saving 5+ hours of manual work per week
- Work with Docker and Kubernetes in Azure environments to build, containerize, and run application services
- Build and consume REST APIs, integrating enterprise systems with Adobe, Microsoft, and Asana using secure authentication and data integration patterns
- Administer MSSQL databases, including backups, indexing, and access permissions
- Write automated unit and integration tests and advocate for consistent quality through code reviews and technical documentation throughout the software development lifecycle
- Participate in Agile processes including sprint planning, demos, and cross-functional collaboration with stakeholders
- Use AI coding tools (Claude, GitHub Copilot, Cursor) daily to write, test, and debug code, accelerating feature development, test generation, and automation scripting while reviewing all output for correctness

Web Designer (Co-op) **Sep 2020 – Dec 2020**

Invest Ottawa *Ottawa, ON*

- Designed, developed, tested, and launched e-commerce websites for three small businesses, establishing online sales capabilities and improving digital presence
- Collaborated with designers, business owners, and marketing teams to translate business requirements into effective digital solutions
- Troubleshot front-end issues and optimized responsive layouts across browsers and screen sizes

EDUCATION

Carleton University **Sep 2024 – Apr 2028 (Expected)**

B.S. Computer Science, AI & ML Stream, Mathematics Minor | Third-year standing | GPA: 11.42/12 *Ottawa, ON*

Algonquin College **May 2019 – Apr 2021**

Computer Programming, Diploma | GPA: 3.92/4 *Ottawa, ON*

PROJECTS

Wealthy, Portfolio Intelligence Platform *Python, React, TypeScript, Claude API*

- Independently building a portfolio analytics platform with a React/TypeScript front end and scheduled Python data pipelines that ingest real-time market data into SQLite, track portfolio performance, and generate daily profit/loss summaries
- Building LLM-powered investment analysis features with the Claude API, combining historical prices, company fundamentals, and technical indicators to generate insights

HintonMarket, Farmers Market Management System *C++, Qt, SQLite*

- Developed a C++ desktop retail management application with inventory, sales, customer, and reporting modules using object-oriented design on a relational SQLite backend
- Designed and integrated the Qt user interface with backend components, implementing input validation, error handling, and debugging practices to deliver a reliable, maintainable application using Git-based version control

VOLUNTEER EXPERIENCE

Khalsa Aid **2024 – 2026**

- Helped package and deliver over 100,000 lbs of food to more than 20 food banks and housing communities through annual food drives

Bluesfest **2019 – 2023**

- Supported festival operations, event logistics, customer service, and fundraising activities